

2 GHz Bandpass Filter Design

Design Calculations — Keysight ADS

Filter Type	Bandpass
Approximation	Chebyshev
Order (n)	3
Centre Frequency	$f_0 = 2 \text{ GHz}$
Passband	~1.885 - 2.115 GHz
Port Impedance	$Z_0 = 50 \text{ Ohm}$

1. Chebyshev Prototype Values

The normalised lowpass prototype element values for a Chebyshev approximation are computed using the recursion formulas below, where L_{Ar} is the passband ripple in dB:

Formulas

$$\beta = \ln(\coth(L_{Ar} / 17.37))$$

$$\gamma = \sinh(\beta / 2n)$$

$$a_k = \sin[(2k - 1) * \pi / 2n] \quad k = 1, 2, 3, \dots, n$$

$$b_k = \gamma^2 + \sin^2(k\pi / n) \quad k = 1, 2, 3, \dots, n$$

$$g_1 = 2 * a_1 / \gamma$$

$$g_k = (4 * a_{(k-1)} * a_k) / (b_{(k-1)} * g_{(k-1)}) \quad k = 2, 3, \dots, n$$

$$g_{(n+1)} = 1 \text{ for } n \text{ odd}$$

$$g_{(n+1)} = \coth^2(\beta / 4) \text{ for } n \text{ even}$$

Resulting Prototype Values

For a 3rd-order Chebyshev filter the g-values map to element types as follows:

$g_1 = 1.5963$	= C_1 (shunt capacitor)
$g_2 = 1.0967$	= L_2 (series inductor)
$g_3 = 1.5963$	= C_3 (shunt capacitor)
$g_4 = 1.0000$	(load termination)

Note: Symmetry $g_1 = g_3$ and $g_4 = 1$ confirms a 3rd-order Chebyshev with equal-ripple passband and matched terminations.

2. Lumped Element Bandpass Filter

The lowpass prototype is transformed to a bandpass filter centred at $f_0 = 2 \text{ GHz}$ through frequency and impedance scaling. The fractional bandwidth Delta is:

$$\Delta = (\omega_2 - \omega_1) / \omega_0$$

where $Z_0 = 50 \text{ Ohm}$.

Scaling Relations

Series arm — from shunt prototype element C_k :

$$L'_k = (C_k * Z_0) / (\omega_0 * \Delta)$$

$$C'_k = \Delta / (C_k * Z_0 * \omega_0)$$

Shunt arm — from series prototype element L_k :

$$L'_k = (\Delta * Z_0) / (\omega_0 * L_k)$$

$$C'_k = L_k / (Z_0 * \Delta * \omega_0)$$

Resulting Lumped Element Values

$L'_1 = L'_3$	63 nH
$C'_1 = C'_3$	0.1004 pF
L'_2	0.365 nH
C'_2	17.34 pF

Topology: Two series LC resonators (L_1 - C_1 and L_3 - C_3) in the main signal path, with a shunt parallel LC resonator ($L_2 \parallel C_2$) to ground at the centre node.

3. Microstrip Coupled-Line Conversion

The lumped BPF is converted to a physically realisable microstrip coupled-line bandpass filter using the **odd/even mode impedance method**. Each coupled-line section has an electrical length of 90 degrees at $f_0 = 2$ GHz.

Admittance Inverter Values

J_k values are derived from the prototype g-values and fractional bandwidth Δ :

$$J_1 / Y_0 = \sqrt{\pi * \Delta / (2 * g_1)}$$

$$J_k / Y_0 = \sqrt{\pi * \Delta^2 / (4 * g_{(k-1)} * g_k)} \quad k = 2, 3, \dots, n$$

$$J_{(n+1)} / Y_0 = \sqrt{\pi * \Delta / (2 * g_n * g_{(n+1)})}$$

Odd / Even Mode Impedance Equations

$$Z_{0e,k} = Z_0 * [1 + J_k * Z_0 + (J_k * Z_0)^2]$$

$$Z_{0o,k} = Z_0 * [1 - J_k * Z_0 + (J_k * Z_0)^2]$$

Substrate Parameters

Material	FR4
Dielectric Constant ϵ_r	4.6
Substrate Thickness H	1.6 mm
Conductor	Copper
Conductor Thickness T	35 μ m
Loss Tangent	0.002

Electrical Length	90 deg at 2 GHz
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Coupled-Line Section Dimensions

Section	Z0e (Ohm)	Z0o (Ohm)	Width (mm)	Spacing (mm)	Length (mm)
1	70.61	39.24	2.290	0.491	20.723
2	56.64	44.77	2.803	1.804	20.255
3	56.64	44.77	2.803	1.804	20.255
4	70.61	39.24	2.290	0.491	20.723

Feed lines (TL1, TL2): W = 2.918 mm, L = 5 mm (50 Ohm microstrip)

Note: Symmetry of sections 1 and 4, and sections 2 and 3, is consistent with the symmetric Chebyshev prototype ($g_1 = g_3$).

4. ADS Optimisation

After the analytical conversion, a **random optimiser** was run in Keysight ADS to fine-tune the coupled-line dimensions and compensate for discontinuity effects not captured by the ideal circuit model.

Optimiser Type	Random
Max Iterations	3326
Goal 1	S(1,1) minimisation in passband
Goal 2	S(2,1) passband flatness
Weight S(1,1)	1
Weight S(2,1)	1 to 2

Post-Optimisation S-Parameter Results

Marker / Frequency	S-Parameters
m5 1.885 GHz	S(1,1) = -3.011 dB S(2,1) = -3.987 dB
m4 1.995 GHz	S(1,1) = -24.253 dB S(2,1) = -0.384 dB
m3 2.115 GHz	S(1,1) = -3.425 dB S(2,1) = -3.448 dB

5. EM Simulation — ADS Momentum (MoM)

Full-wave EM simulation was performed using **ADS Momentum (MoM Microwave)** to validate the physical layout, accounting for conductor losses, fringing fields, and substrate parasitics.

Frequency Plan

Solver	Momentum Microwave (MoM)
Sweep Type	Linear

Fstart	1 GHz
Fstop	3 GHz
Points	200
Step	~10 MHz

Substrate Stackup

Top dielectric	AIR
Conductor 1 (cond)	Copper 35 um
Dielectric	FR4 1.6 mm
Conductor 2 (cond2)	Copper 35 um
Bottom dielectric	AIR

EM Simulation S-Parameter Results

Marker / Frequency	S-Parameters
m5 1.884 GHz	S(1,1) = -1.063 dB S(2,1) = -29.558 dB
m4 1.995 GHz	S(1,1) = -7.527 dB S(2,1) = -9.358 dB
m3 2.116 GHz	S(1,1) = -7.536 dB S(2,1) = -6.666 dB

Note: The EM response shows degradation compared to the circuit-level simulation. This is expected: Momentum captures conductor losses, fringing fields, and substrate effects that the ideal coupled-line model does not include. The passband shift and insertion loss increase are primarily attributed to FR4's loss tangent (0.002) at 2 GHz and mesh resolution. A finer mesh and substrate refinement would improve accuracy.

6. Results Summary

Stage	Centre Freq	S(2,1) at f0	Passband BW
Lumped Element	~2 GHz	~0 dB	~230 MHz
Microstrip (opt.)	1.995 GHz	-0.384 dB	~230 MHz
EM Simulation	1.995 GHz	-9.358 dB	degraded